

Digital Signal Processing Assignment Report

Course: EEE-3207

Topic: Analyzing Audio Signals of Distinctive Words Using MATLAB

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Submission Date: 08/07/2025

1. Objective

The objective of this assignment was to analyze Bangla speech signals, specifically the words "chaar" and "tiin", using MATLAB. The analysis involved:

- Audio signal preprocessing
- Frequency domain analysis using FFT
- Vowel detection using cross-correlation

2. Audio Processing

2.1 Audio Recording

Three recordings were created using different voices pronouncing the two words "chaar" and "tiin". Additionally, individual vowels "aa" and "ii" were recorded as aaa.mp3 and iii.mp3 respectively.

2.2 Preprocessing Steps

- Converted stereo audio to mono
- Normalized amplitude
- Extracted word segments manually using time markers:
 - `chaar = audio(round(0.023*fs):round(0.4154*fs));`
 - `tiin = audio(round(0.5364*fs):round(0.93*fs));`

3. Fourier Analysis

3.1 Applying FFT

We applied the Fast Fourier Transform (FFT) to the word signals to examine their spectral content. Only the first half of the FFT (non-redundant part) was analyzed.

3.2 Frequency Spectra and Peaks

- "chaar":
 - Peak frequency: **586.11 Hz** (related to vowel "aa")
- "tiin":
 - Peak frequency: **233.72 Hz** (related to vowel "ii")

3.3 Spectral Plots

Plots were generated for each word using MATLAB's plot() function and saved in the plots/ folder.

4. Cross-Correlation Analysis

4.1 Purpose

To detect vowel presence in each word using similarity matching via cross-correlation.

4.2 Method

- Used xcorr() to compare:
 - ❖ aaa.mp3 ("aa") with chaar and tiin
 - ❖ iii.mp3 ("ii") with chaar and tiin

4.3 Results

Comparison	Peak Correlation
aa vs chaar	53.58
aa vs tiin	47.50
ii vs chaar	64.73
ii vs tiin	77.07

4.4 Interpretation

- "aa" matches better with "chaar" than "tiin"
- "ii" has the strongest correlation with "tiin"
- This confirms that vowel detection via cross-correlation is effective

4.5 Correlation Plots

Each correlation array was plotted. These are stored in the plots/ folder for visual inspection.

5. Reporting Results

5.1 Fourier Spectra Summary

- "chaar": Strong energy around **586.11 Hz**
- "tiin": Strong energy around **233.72 Hz**

5.2 Vowel Frequencies

- "aa": -586.11 Hz (open vowel)
- "ii": -233.72 Hz (front-high vowel)

5.3 Cross-Correlation Summary

- Peak correlation values confirm vowel-to-word relationships:
 - "aa" with "chaar": 53.58
 - "ii" with "tiin": 77.07

7. Conclusion

This assignment provided hands-on experience with practical signal processing techniques such as:

- FFT for frequency-domain analysis
- Cross-correlation for speech pattern detection

The vowel-frequency correlation successfully differentiated the words "chaar" and "tiin", showing how MATLAB can be used in basic speech recognition tasks.

End of Report